

13:5-Chain Rules for Functions of Several Variables

Dr. Md. Abul Kalam Azad
Associate Professor, Mathematics
NSc,IUT

Objectives

- Use the Chain Rules for functions of several variables.
- Find partial derivatives implicitly.

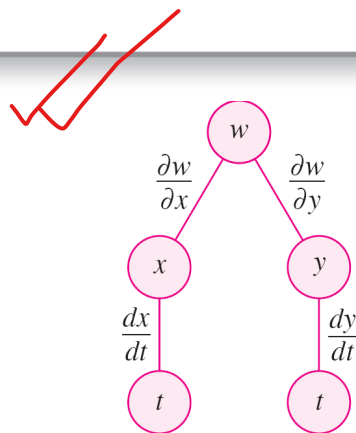
Chain Rules for Functions of Several Variables

THEOREM 13.6 Chain Rule: One Independent Variable

Let $w = f(x, y)$, where f is a differentiable function of x and y . If $x = g(t)$ and $y = h(t)$, where g and h are differentiable functions of t , then w is a differentiable function of t , and

$$\frac{dw}{dt} = \frac{\partial w}{\partial x} \frac{dx}{dt} + \frac{\partial w}{\partial y} \frac{dy}{dt}.$$

The Chain Rule is shown schematically in Figure 13.39.



Chain Rule: one independent variable w is a function of x and y , which are each functions of t . This diagram represents the derivative of w with respect to t .

Figure 13.39

Example 1 – Chain Rule: One Independent Variable

- Let $w = x^2y - y^2$, where $x = \sin t$ and $y = e^t$. Find dw/dt when $t = 0$.

- Solution:**

By the Chain Rule for one independent variable, you

have

$$\frac{dw}{dt} = \frac{\partial w}{\partial x} \frac{dx}{dt} + \frac{\partial w}{\partial y} \frac{dy}{dt}$$

$$= 2xy(\cos t) + (x^2 - 2y)e^t$$

Example 1 – Solution

$$= 2(\sin t)(e^t)(\cos t) + (\sin^2 t - 2e^t)e^t$$

$$= 2e^t \sin t \cos t + e^t \sin^2 t - 2e^{2t}.$$

When $t = 0$, it follows that

$$\frac{dw}{dt} = -2.$$

Chain Rules for Functions of Several Variables

- The Chain Rule in Theorem 13.6 can provide alternative techniques for solving many problems in single-variable calculus. For instance, in Example 1, you could have used single-variable techniques to find dw/dt by first writing w as a function of t ,

$$\begin{aligned}w &= x^2y - y^2 \\&= (\sin t)^2(e^t) - (e^t)^2 \\&= e^t \sin^2 t - e^{2t}\end{aligned}$$

and then $\frac{dw}{dt} = 2e^t \sin t \cos t + e^t \sin^2 t - 2e^{2t}$

Chain Rules for Functions of Several Variables

- The Chain Rule in Theorem 13.6 can be extended to any number of variables. For example, if each x_i is a differentiable function of a single variable t , then for

$$w = f(x_1, x_2, \dots, x_n)$$

- you have

$$\left[\frac{dw}{dt} = \frac{\partial w}{\partial x_1} \frac{dx_1}{dt} + \frac{\partial w}{\partial x_2} \frac{dx_2}{dt} + \dots + \frac{\partial w}{\partial x_n} \frac{dx_n}{dt} \right]$$

Chain Rules for Functions of Several Variables

THEOREM 13.7 Chain Rule: Two Independent Variables

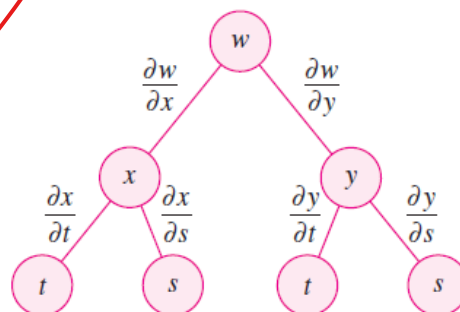
Let $w = f(x, y)$, where f is a differentiable function of x and y . If $x = g(s, t)$ and $y = h(s, t)$ such that the first partials $\partial x/\partial s$, $\partial x/\partial t$, $\partial y/\partial s$, and $\partial y/\partial t$ all exist, then $\partial w/\partial s$ and $\partial w/\partial t$ exist and are given by

$$\frac{\partial w}{\partial s} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial s}$$

and

$$\frac{\partial w}{\partial t} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial t}$$

The Chain Rule is shown schematically in Figure 13.41.



Chain Rule: two independent variables

Figure 13.41

Example 4 – The Chain Rule with Two Independent Variables

- Use the Chain Rule to find $\partial w / \partial s$ and $\partial w / \partial t$ for $w = 2xy$ where $x = s^2 + t^2$ and $y = s/t$.

- Solution:**

Using Theorem 13.7, you can hold t constant and differentiate with respect to s to obtain

$$\begin{aligned}\frac{\partial w}{\partial s} &= \frac{\partial w}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial s} \\ &= 2y(2s) + 2x\left(\frac{1}{t}\right)\end{aligned}$$

Example 4 – Solution

cont'd

$$= 2\left(\frac{s}{t}\right)(2s) + 2(s^2 + t^2)\left(\frac{1}{t}\right) \quad \text{Substitute } (s/t) \text{ for } y \text{ and } s^2 + t^2 \text{ for } x.$$

$$= \frac{4s^2}{t} + \frac{2s^2 + 2t^2}{t}$$

$$= \frac{6s^2 + 2t^2}{t} \cdot \quad \text{no final value can be reached}$$

Similarly, holding s constant gives

$$\frac{\partial w}{\partial t} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial t}$$

Example 4 – Solution

cont'd

$$= 2y(2t) + 2x\left(\frac{-s}{t^2}\right)$$

$$= 2\left(\frac{s}{t}\right)(2t) + 2(s^2 + t^2)\left(\frac{-s}{t^2}\right) \quad \text{Substitute } (s/t) \text{ for } y \text{ and } s^2 + t^2 \text{ for } x.$$

$$= 4s - \frac{2s^3 + 2st^2}{t^2}$$

$$= \frac{4st^2 - 2s^3 - 2st^2}{t^2}$$

$$= \frac{2st^2 - 2s^3}{t^2}.$$

Chain Rules for Functions of Several Variables

- The Chain Rule in Theorem 13.7 can also be extended to any number of variables. For example, if w is a differentiable function of the n variables $x_1, x_2, \dots, x_n$ where each x_i is a differentiable function of the m variables $t_1, t_2, \dots, t_m$, then for $w = f(x_1, x_2, \dots, x_n)$ you obtain the following.

$$\left[\frac{\partial w}{\partial t_1} = \frac{\partial w}{\partial x_1} \frac{\partial x_1}{\partial t_1} + \frac{\partial w}{\partial x_2} \frac{\partial x_2}{\partial t_1} + \dots + \frac{\partial w}{\partial x_n} \frac{\partial x_n}{\partial t_1} \right]$$

$$\frac{\partial w}{\partial t_2} = \frac{\partial w}{\partial x_1} \frac{\partial x_1}{\partial t_2} + \frac{\partial w}{\partial x_2} \frac{\partial x_2}{\partial t_2} + \dots + \frac{\partial w}{\partial x_n} \frac{\partial x_n}{\partial t_2}$$

$$\vdots$$

$$\frac{\partial w}{\partial t_m} = \frac{\partial w}{\partial x_1} \frac{\partial x_1}{\partial t_m} + \frac{\partial w}{\partial x_2} \frac{\partial x_2}{\partial t_m} + \dots + \frac{\partial w}{\partial x_n} \frac{\partial x_n}{\partial t_m}$$

Implicit Partial Differentiation

- This section concludes with an application of the Chain Rule to determine the derivative of a function defined *implicitly*.
- ✓ Let x and y be related by the equation $F(x, y) = 0$, where $y = f(x)$ is a differentiable function of x . To find dy/dx , you could use the techniques discussed in Section 2.5. You will see, however, that the Chain Rule provides a convenient alternative. Consider the function $w = F(x, y) = F(x, f(x))$.
- You can apply Theorem 13.6 to obtain

$$\left[\frac{dw}{dx} = F_x(x, y) \frac{dx}{dx} + F_y(x, y) \frac{dy}{dx} \right]$$

Implicit Partial Differentiation

THEOREM 13.8 Chain Rule: Implicit Differentiation

If the equation $F(x, y) = 0$ defines y implicitly as a differentiable function of x , then

$$\frac{dy}{dx} = -\frac{F_x(x, y)}{F_y(x, y)}, \quad F_y(x, y) \neq 0.$$

If the equation $F(x, y, z) = 0$ defines z implicitly as a differentiable function of x and y , then

$$\frac{\partial z}{\partial x} = -\frac{F_x(x, y, z)}{F_z(x, y, z)} \quad \text{and} \quad \frac{\partial z}{\partial y} = -\frac{F_y(x, y, z)}{F_z(x, y, z)}, \quad F_z(x, y, z) \neq 0.$$

✓ Example 6 – Finding a Derivative Implicitly

- Find dy/dx for $y^3 + y^2 - 5y - x^2 + 4 = 0$.

- Solution:**

Begin by letting

$$F(x, y) = y^3 + y^2 - 5y - x^2 + 4.$$

Then

$$F_x(x, y) = -2x \quad \text{and} \quad F_y(x, y) = 3y^2 + 2y - 5.$$

- Using Theorem 13.8, you have

$$\frac{dy}{dx} = -\frac{F_x(x, y)}{F_y(x, y)} = \frac{-(-2x)}{3y^2 + 2y - 5} = \frac{2x}{3y^2 + 2y - 5}.$$

Suggested Problems

Exercise 13.5: **11,17,22,25,29.**

Thanks a lot ...